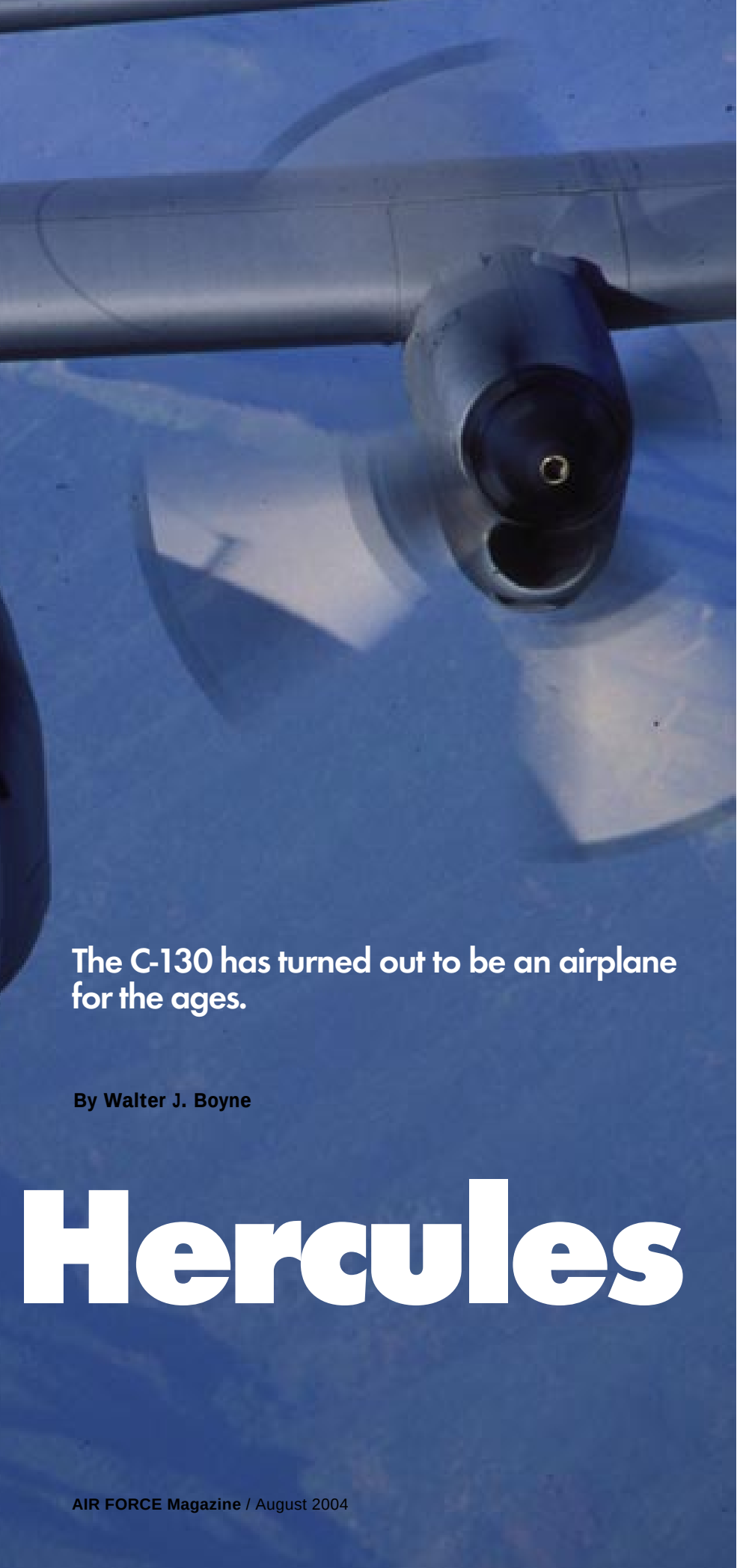
Photo by Ted Carlson

The Immortal



The C-130 has turned out to be an airplane for the ages.

By Walter J. Boyne

Hercules

IT WAS exactly 2:45 p.m. on Aug. 23, 1954, in Burbank when the prototype of a brand-new Air Force transport slowly took off into the smoggy skies over southern California. As they watched it climb out of sight, Lockheed officials dared to hope that the Air Force might buy as many as 100 of these new aircraft.

Talk about answered prayers.

Even the airplane's most ardent supporters could not have foreseen that the humble C-130 would enjoy the longest continuous production run of any military aircraft in history. USAF snapped up those first 100 airplanes and just kept on going—for decades, with no sign of stopping anytime soon.

Still under its original type certificate, the Hercules remains in production 50 years after that maiden flight. Lockheed has delivered 2,262 C-130s to some 60 countries. Even today, the aerospace giant enjoys a healthy backlog; it is working off firm orders for 71 of the latest variant—the C-130J.

No one would have believed that an aircraft designed as a workhorse "trash hauler" would undertake such a variety of missions. It has dropped bombs, supplies, and paratroops, jammed electronic transmissions, fought fires, tracked icebergs, flown in hurricanes, hauled a live whale and camels, carried Muslims to Mecca, taken Ethiopian Jews to Israel, and even landed on an aircraft carrier.

Four C-130s were used to form the Four Horseman aerial demonstration team. The "Herk" has flown to most countries and every continent. It has landed in the Arctic and Antarctic. For the last 50 years, it has usually been among the first airplanes to arrive at a trouble spot.

It has served as a gunship, tanker, bomber, and drone mother ship. It has supported psychological warfare, special operations, electronic intelligence, command and control, and humanitarian rescue and relief.

The Air Force experience with cargo aircraft early in the Korean War convinced senior leaders that USAF needed a more capable trans-

Lockheed built the prototype C-130s in Burbank, Calif., but moved production to Marietta, Ga. (above), where it is today. In 1951, B-47 production (in background) was ending.

port. The Fairchild C-119 proved to be marginally more effective (and less reliable) than the Douglas C-47s and Curtiss C-46s from World War II.

Birth of a Program

So it was that, on Feb. 2, 1951, the Air Force put forth a general operational requirement that called for a huge advance in cargo aircraft capability. Lockheed, Boeing, Douglas, and Fairchild were invited to compete for the contract.

All of the specifications for range, load, and operating conditions were formidable (see box at right). The most daunting of these, however, was the requirement that the airplane have the ability to fly with a full load with one engine out. In the past, twin-engine aircraft, especially those operating out of short fields in forward areas, usually did not survive the loss of an engine on a heavy-weight takeoff.

Willis M. Hawkins, then head of preliminary design for Lockheed, put together a team of veteran Lockheed engineers that included Eugene Frost, Art Flock, and Dick Pulver, all of whom had worked together on other projects. Notably absent from the team was Lockheed's most well-known engineer, Clarence "Kelly" Johnson, who was deeply involved in the F-104 project.

In June 1951, the Hawkins team completed its proposal for what Lockheed called the Model 82 aircraft and took it to Hall L. Hibbard, Lockheed's chief of engineering. The entire proposal was about three-quarters of an inch thick.

Hibbard asked, "Has Kelly seen this?" When Hawkins said no, the group asked Johnson to come review it.

Johnson went through the drawings, glanced at the model Hawkins had provided, and then declared to Hibbard, "If you send that in, you'll destroy Lockheed."

Johnson's reaction to the C-130 was based in part on aesthetics. Lockheed was known for building beautiful aircraft, from the early Vega through the P-38 and Constellation. The Hercules, as the new aircraft would be called, was not exactly beautiful.

Fortunately Hawkins persisted and Hibbard backed him. They knew that, despite its plain looks, the C-130 was a radically advanced transport, using four T56 turboprop engines and featuring a completely pressurized cargo compartment.

Form had followed function, and the heart of the aircraft was the huge 4,500 cubic foot cargo area that duplicated the volume of the standard American railroad boxcar. The use of a high wing and rugged dual-tandem wheel landing gear system, mounted in stub-like fairings outside the fuselage, improved its short, rough-field capabilities.

On to Georgia

Lockheed won the competition, and construction of two prototypes began in Burbank.

The first flight was staged by the second (Serial No. 53-3397) of the two prototypes. It was piloted by Stanley Beltz and Roy Wimmer, with Jack Real as flight-test engineer and Dick Stanton as flight engineer. Johnson flew in a chase airplane. After a satisfying 61-minute flight, the YC-130 landed at Edwards AFB, Calif., where it awaited further tests.

The new aircraft exceeded all goals, cruising faster, climbing higher, and landing on less runway than required in any of the Air Force specifications. The C-130 had a maximum payload of 40,000 pounds, thanks in part to the weight control measures that kept the airframe

In the Beginning ...

Here is an excerpt from the original General Operational Requirement for Cargo Aircraft, issued in 1951.

The aircraft must be able to:

1 Carry 92 infantrymen or 64 paratroopers on a mission with a combat radius of 1,100 nautical miles, or, alternatively, a 30,000-pound cargo more than 960 miles.

2 Operate from short unprepared airstrips of clay, sand, or humus soil.

3 Slow down to 125 knots for paradrops and even slower for assault landings.

4 Have both a rear ramp operable in flight for heavy equipment and side doors for paratroop drops.

5 Handle bulky and heavy equipment, including bulldozers, artillery pieces, and trucks.

6 Fly with one engine out.

This C-130A is not stuck: It's taxiing in sandy soil at Eglin AFB, Fla., in 1959, demonstrating its capability to operate on unimproved strips. The early C-130s featured three-bladed props and a "Roman nose," later extended for a radar.

down to 108,000 pounds, 5,000 less than anticipated.

When the Air Force issued a contract for the first seven production aircraft, Lockheed decided to move the program to Marietta, Ga., where Lockheed had built Boeing B-47s under license. B-47 production was coming to a close, and the C-130 program was perfectly timed to pick up the slack.

Shortly after the successful first flight, the Air Force increased its production order from seven to 75 airplanes.

Production went smoothly at the Georgia plant, despite a mishap to the first production aircraft (53-3129), which suffered a major in-flight fire in its No. 2 engine nacelle on its third flight. The aircraft landed without further incident. The left wing was replaced. (This specific aircraft was subsequently modified to become an AC-130A gunship and saw service in the Vietnam War. It is now at the USAF Armament Museum, Eglin AFB, Fla.)

The most significant engineering change stemmed from the unsatisfactory operation of the turboelectric propeller. At one point, 50 completed C-130s could not be delivered because no decision had been made about which propeller to use. Finally, a new hydraulically operated propeller was selected, and it mated perfectly with the engine.

The Hercules entered the Tactical Air Command (TAC) fleet Dec. 9, 1956, with the delivery of 55-0023 to the 463rd Troop Carrier Wing at Ardmore AFB, Okla. Crews were delighted, for the aircraft was far nimbler than the C-119s. It also had surplus takeoff power.

Deliveries to TAC continued on a regular basis, and two C-130 units, the 463rd and the 314th TCW, Stewart AFB, Tenn., formed an important part of the Composite Air Strike Force.

Wherever the C-130 went, it brought new standards of performance along with vastly improved comfort and reliability. C-130s were called on to fly troops, weapons, and ammunition to trouble spots around the world. One early case occurred in July 1958, when turmoil in Iraq caused Lebanese President Camille Chamoun to seek a US troop presence in his country. An 11-day airlift brought eight million pounds of equipment into Lebanon.

With hundreds of similar incidents to come, the ability of the C-130 to move troops and equipment directly to a crisis zone became an essential part of US military and diplomatic power.

Hercules Down

The first combat loss of the C-130 occurred Sept. 2, 1958, when Soviet pilots flying MiG-17s shot down a United States Air Force C-130A-II signals intelligence platform over Soviet Armenia. All 17 crew members were killed.

Many more losses were to occur in Vietnam, where the C-130 formed the backbone of the airlift system. About 50 C-130s were lost in combat between 1965 and 1972. Few if any of the losses stemmed from accidents.

In Vietnam, no other theater airlifter could match the capacity or the versatility of the Hercules. The C-130s not only underpinned the tight logistics network throughout Southeast Asia, but also saw the war up close, bringing troops and equipment directly to front-line action within range of enemy guns. The C-130 radar permitted it to operate in a

Staff photo by Capt. Mike Perini

Ski-equipped C-130s provided a lifeline to the facilities along the DEW Line in Greenland (above) and resupplied scientific teams at the poles. The Herk quickly gained a reputation as the "go anywhere" airplane.

The rugged Hercules was first adapted to duty as a gunship in Vietnam. Bristling with sensors and cannons, AC-130 Spectres have rained down fire on enemies in every US action since 1967.

much wider range of weather, and this capability led logically to it being employed later as a gunship.

In its best-known Vietnam exploit, the C-130 fleet frustrated North Vietnamese Gen. Vo Nguyen Giap's efforts to trap American forces at Khe Sanh. Giap wanted to score a significant propaganda victory by capturing a large number of prisoners, and, to that end, he sent two regular North Vietnamese Army divisions to surround the 6,000 Marines defending the Khe Sanh garrison.

During the 70-day siege in early 1968, 92 percent of all supplies were brought in by C-130s. Other elements of American airpower, including close air support, helped the Marines resist, but it was the C-130s that kept them supplied and operating.

The C-130s would land at Khe Sanh after a steep approach and off-load cargo as swiftly as possible. When it was too dangerous to land, the C-130s would achieve the objective by using the low-altitude parachute extraction system technique (LAPES). When neither landing nor LAPES was possible, the C-130s would air-drop their cargoes.

In every instance, the transports were vulnerable to enemy fire.

The first AC-130 Spectre gunship commenced operations from Nha Trang in September 1967. It was so successful that the Air Force built 28 more. The effect of the Spectre's firepower was startling. In one minute, its 20 mm gun could saturate an area the size of a football field. The last 11 AC-130s were equipped with the 105 mm howitzer.

The AC-130 performed spectacularly in the April to June 1972 battle for An Loc.

The C-130 also served as a bomber. In Operation Commando Vault, C-130s flew hundreds of bombing sorties to clear a jungle area for use by helicopters. During the Tet Offensive in early 1968, C-130s bombed enemy troops with improvised bombs. The Herk can now handle the Massive Ordnance Air Blast (MOAB) bomb, more colloquially known as the "Mother of All Bombs."

Continuous improvement of the aircraft over the years, particularly the increase in performance resulting from the use of new and more powerful engines, made it attractive for a wide range of roles.

Seventy, So Far

There have been at least 70 C-130 variants. Some were built in small numbers for tasks that differed only slightly from the routine, while others were built for highly specialized tasks, far removed from the concept of carrying troops and cargo from Point A to Point B. Some aircraft, after having fulfilled the new duties of a specific mission, were converted back to standard C-130 transport configuration.

Gathering signals intelligence was one of the first additional missions, and 10 C-130A-II-LM aircraft were modified for use by the 7407th Combat Support Wing. This tradition has been expanded in today's EC-130 counterparts.

The now-retired EC-130 ABCCC (Airborne Battlefield Command and Control Center) was an effective supplement to the larger E-3 Airborne Battlefield Warning and Control System aircraft.

The EC-130 Commando Solo is used in psychological warfare, carrying such powerful radio and television broadcasting equipment that it literally becomes the one voice

In Vietnam in 1968, the Herk resupplied the besieged Marine garrison at Khe Sanh. Exposed to enemy fire, C-130s often performed pallet insertion of desperately needed materiel, as seen here.

that can be seen and heard in its broadcast area.

The Hercules offered the Marine Corps a chance to obtain a suitable aerial tanker for its aircraft. The first of these, originally designated GV-1s but subsequently redesignated KC-130F, entered service in 1960. One of the most remarkable capabilities of the Hercules was the in-flight refueling of helicopters. This not only helped choppers conduct conventional missions but also opened a broad new area of helicopter tactics.

The C-130 was especially valuable for the search and rescue role, with HC-130H aircraft acting both as command and control aircraft and tanker. The Air Force uses the HC-130P version currently for combat search and rescue.

Some Hercules were modified to become MC-130E Combat Talon I aircraft used for special operations. They have in-flight refueling receptacles and infrared detection equipment, and some used to carry Fulton rescue gear. The follow-on MC-130H Combat Talon II is a new-build aircraft with additional equipment. The MC-130P Combat Shadow is dedicated to long-distance, clandestine, low-level missions into denied areas to provide air refueling to special operations forces helicopters.

Lockheed Martin photo by John Rossino

The next generation C-130J has a six-bladed prop, more powerful engines, and a digital cockpit. With better performance and unmatched flexibility, the Herk will be around for decades to come.

Long Takeoff at Tan Son Nhut

On April 29, 1975, the fall of Saigon was imminent, and nearby Tan Son Nhut Air Base was under heavy fire. South Vietnamese Air Force officer Tinh Nguyen saw a single C-130A taxiing out. The cargo ramp was still open, with desperate people clambering on board. Nguyen joined them.

At the end of the runway, the cargo door finally closed, and the pilot powered up. The overweight Hercules slowly ran down the 9,000-foot runway, finally staggering off the ground at the end of the 1,000-foot overrun. The C-130 stayed in ground effect until it gained enough speed to begin a shallow climb.

The airplane was at least 20,200 pounds overweight, as it carried 452 people, including 33 crowded into the flight deck.

After a flight lasting nearly four hours, the C-130 landed at U Tapao RTAB, Thailand. When Nguyen got out, he looked at the C-130 and vowed that he would someday work for the company that built such a remarkable airplane.

Today, he does just that. Nguyen works at Lockheed Martin in Marietta, Ga., where he is a specialist in defensive systems. The aircraft that carried him and 451 others to safety may now be found as the gate guardian at Little Rock AFB, Ark.

Unique Roles

In addition to broad missions as outlined above, many Hercules were used for unique roles that sometimes required only a few aircraft. These versions included weather reconnaissance aircraft (WC-130), a ski-equipped version (LC-130) for use in both the Arctic and Antarctic, "TACAMO" (Take Charge and Move Out EC-130G) that linked the National Command Authority to submarines on patrol, and a satellite recovery version (NC-130H). Perhaps the most dramatic of all was the YMC-130H. Under a project called Credible Sport, this specially equipped C-130 was to participate in the ill-fated 1980 attempt to rescue hostages held by Iran. The YMC-130H was to make extremely short field landings and takeoffs using booster rockets and retrorockets. One example of the three YMC-130Hs may be seen at the Museum of Aviation, Robins AFB, Ga.

The first among the many foreign users of the C-130 was the Royal Australian Air Force, which obtained 10 C-130As beginning in 1957. The United Kingdom purchased the most aircraft, 66, while Saudia Arabia is second, with 50.

The Israeli Defense Force received 12 C-130s during the October 1973 war, and they were pressed into service, taking ammunition directly to front-line units. The Israeli C-130s performed as flying trucks, following tanks into battle, "S" turning to maintain position, and landing on a spot to deliver ammunition and fuel directly to the armored forces.

The Hercules has been around so long that one tends to take it for granted. Observers see the C-130 operating effectively 50 years after its first flight and think it's perfectly routine. The same observations will probably be made decades hence, when, almost certainly, later models of the C-130 will be going strong. ■

Walter J. Boyne, former director of the National Air and Space Museum in Washington, D.C., is a retired Air Force colonel and author. He has written more than 400 articles about aviation topics and 29 books, the most recent of which is The Two O'Clock War: The 1973 Yom Kippur Conflict and the Airlift That Saved Israel. *His most recent article for* Air Force *Magazine, "Von Karman's Way," appeared in the January issue.*